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ZOO501-Developmental-Biology

Solved Questions FINAL TERM (PAST PAPERS)

1. Exon Intron?

Both **introns** and **exons** are the genetic material. But the essential **difference between** the two is that one is made into protein and the other is not. DNA molecules are made up of small molecules that are called nucleic acids. An **exon** is a sequence of nucleic acids that are represented **in the** mRNA molecule.

2. How to avoid polyspermy?

prevent the entry of more than one sperm into the egg. The sea urchin egg has two mechanisms to **avoid polyspermy**: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a slower reaction, caused by the exocytosis of the cortical granules

3. Defect of neurulation?

Neural tube **defects** (NTDs) are severe congenital malformations affecting 1 in every 1000 pregnancies. 'Open' NTDs result from failure of primary **neurulation** as seen in anencephaly, myelomeningocele (open spina bifida) and craniorachischisis.

4. Meroblastic cleavage?

meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg. These cells form a blastoderm.

5. Fraternal and Identical twin?

Twins are two offspring produced by the same pregnancy. **Twins** can be either **monozygotic**('**identical**'), meaning that they develop from one zygote, which splits and forms two embryos, or dizygotic ('**fraternal**'), meaning that each **twin** develops from a separate egg and each egg is fertilized by its own sperm cell.

6. What is blastocyst?

Blastocyst: A thin-walled hollow structure in early embryonic development that contains a cluster of cells called the inner cell mass from which the embryo arises.

7. How acrosomal reaction initiate in sea urchin?

In the **sea urchin**, contact with egg jelly **initiates** the **acrosome reaction**, which is a calcium-mediated process. ... The **acrosomal** vesicle (green) fuses with the plasma membrane, releasing enzymes from the tip of the sperm that aid digestion of egg jelly.

8. How genetic material exchange in paramecium?

The micronucleus is the generative, or germline nucleus, containing the **genetic material** that is passed along from one generation to the next. **Paramecium** reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis.

9. What is fate of metencephalon?

The **fate** of the brain vesicles is determined by the combination of transcription factors. Signals from the secondary organizer, the anterior neural ridge and the mesencephalon–**metencephalon** boundary (isthmus), regulate expression of the transcription factors in the adjacent region and organize its **fate**.

10. Function of Placenta?

The **placenta** acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other waste products. It metabolises a number of substances and can release metabolic products into maternal and/or fetal circulations.

11. What is campomelic dysplasia?

Campomelic dysplasia is a severe disorder that affects development of the skeleton, reproductive system, and other parts of the body. This condition is often life-threatening in the newborn period. The term "**campomelic**" comes from the Greek words for "bent limb."6 days ago

12. Name 4 level gene regulation? Level of gene regulation?

13. Two different pathways of neural crest cell? (10)

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.

The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome

14. Myogenesis (10)

Myogenesis: The Development of Muscle Paracrine factors instruct the Myotome cells to become muscles by inducing them to synthesize the MyoD protein. Factors from the surrounding environment induce the Pax3 transcription factor. In the absence of other inhibitory transcription factors (such as those found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, Myf5 and MyoD. MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins (sometimes also referred to as the MyoD family). The proteins of this family all bind to similar sites on the DNA and activate muscle-specific genes. MyoD also directly activates its own gene. While Pax3 is found in several other cell types, the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell. Transfection of genes encoding any of these myogenic proteins into a wide range of cultured cells converts those cells into muscle

Myogenesis is the formation of **muscular** tissue, particularly during embryonic **development**. **Muscle** fibers form from the fusion of myoblasts into multi-nucleated fibers called myotubes. ... In this stage, calcium ions are critical for **development**.

15. Primary and Secondary neurulation?

In **primary** neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.
In **secondary** neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary

16. Cortical granule reaction?

Cortical granules are membrane-bound structures containing proteolytic enzymes. However, whereas each sperm contains one acrosomal vesicle, each sea urchin egg contains approximately 15,000 cortical granules

17. Embryonic Shield?

Also called **embryonic shield**. in the early **embryo** of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the **embryo** begins to differentiate.

18. Slow block polyspermy?

The **slow block** to **polyspermy** begins within 10 seconds of fusion of the sperm and egg plasma membranes. A compound called inositol triphosphate (IP3) causes the release of Ca++ from intracellular stores in the egg endoplasmic reticulum.

19. Osteogenesis?

Osteogenesis imperfecta (OI), also known as brittle bone disease, is a group of genetic disorders that mainly affect the bones. It results in bones that break easily.

20. Types of Cleavage?

the entire cell is divided equally. Four major holoblastic **cleavage types** can be observed in general: radial, spiral, bilateral, and rotational. Egg cells that have larger quantities of yolk undergo meroblastic **cleavage** after fertilization, in which only a portion of the zygote undergoes **cleavage**.

21. Response from egg to sperm?

The responses of the egg to the sperm can be divided into "early" responses, which occur within seconds of the cortical reaction, and "late" responses, which take place several minutes after fertilization begins.

22. Blastocoel function?

The **blastocoel** probably serves two major **functions** in frog embryos: (1) it permits cell migration during gastrulation, and (2) it prevents the cells beneath it from interacting prematurely with the cells above it.

23. Tunicate gastrulation?

Tunicates, like sea urchins and vertebrates, follow the deuterostome pattern of **gastrulation** in which the blastopore becomes the anus.**Tunicate gastrulation** is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm

24. Protozoans? (2)

is an informal term for single-celled eukaryotes, either free-living or parasitic, which feed on organic matter such as other microorganisms or organic tissues and debris.

25. Stereoblastula? (2)

Blastulae produced by radial cleavage have no blastocoel and are called stereoblastulae.

26. Neural Tube Close? (2)

The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births.

27. Function of Enhancer? (3)

Enhancer function underlies regulatory processes by which cells establish patterns of gene expression. Recent results suggest that **enhancers** are specified by particular chromatin marks in pluripotent cells, which may be modified later in development to alter patterns of gene expression and cell differentiation choices.

28. Composition of hyaline membrane in Sea Urchin? (3)

The **hyaline** layer is actually made up of two layers, an outer lamina made primarily of hyalin protein and an inner lamina composed of fibropellin proteins

29. Blastocoel Function? (5)

The **blastocoel** probably serves two major **functions** in frog embryos: (1) it permits cell migration during gastrulation, and (2) it prevents the cells beneath it from interacting prematurely with the cells above it.

30. Blastomere cleavage? (5)

Production of a multicellular organism, many cells must be produced from the single-celled zygote. This task is accomplished by cleavage, a series of consecutive cell divisions. Cells produced during cleavage are called blastomeres.

31. Primary and Secondary Neural tube?

The **neural** plate folds in upon itself to form the **neural tube**, which will later differentiate into the spinal cord and the brain, eventually forming the central nervous system. Different portions of the **neural tube** form by two different processes, called **primary and secondary** neurulation, in different species.

32. What is Blastocyst?

A mammalian blastula in which some differentiation of cells has occurred.

33. How acrosomal reaction initiate in Sea Urchin?

The acrosomal reaction in sea urchins is initiated by contact of the sperm with the egg jelly. Contact with egg jelly causes the exocytosis of the sperm's acrosomal vesicle and the release of proteolytic enzymes that can digest a path through the jelly coat to the egg surface.

34. What is Fete metacephone?

35. What is meroblastic Cleavage?

Depending mostly on the amount of yolk in the egg, **cleavage** can be holoblastic (complete or total) or **meroblastic** (partial). In **meroblastic cleavage**, only a portion of the egg divides. This is because of the large amount of yolk in the egg cell that the **cleavage**furrows do not penetrate the yolk. ... super ficiall **cleavage**.

36. Write 2 major difference pathway of nueral crest cells?

The trunk neural crest, whose cells take one of two major pathways.

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly. The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome. Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.

The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut. The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.

The cardiac neural crest is located between the cranial and trunk neural crests. The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

37. Cortical granule reaction? Cortical reaction?

The cortical reaction is a process initiated during fertilization by the release of cortical granules from the egg, which prevents polyspermy, the fusion of multiple sperm with one egg.

38. Cleavage pattern in sea urchin, nematode and tunicate?

In all four invertebrates described here, **cleavage** is holoblastic. In the **sea urchin**,**cleavage** is radial; in the snail, spiral; in the **tunicate**, bilateral; and in the **nematode**, rotational. ... In some, such as the **sea urchin** and **tunicate**, the axes are established at fertilization through determinants in the egg cytoplasm.

39. How secondary hypoblast form in avians?

Gastrulation of the Avian Embryo Shortly thereafter, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

40. Difference in cotice and nuclei?

41. Larval Stage of Frog?

A tadpole (also called a pollywog) is the **larval stage** in the life cycle of an amphibian, particularly that of a **frog** or toad. They are usually wholly aquatic, though some species have tadpoles that are terrestrial.

42. Commitment Strategy?

43. What happened neural tube fail to fuse?

Neural tube defects—Spina bifida, Anencephaly and Encephalocele. ... The **neural tube** goes on to develop into your baby's brain and spinal cord. When the **neural tube** does not **fuse** together properly a problem with the development of the spine or brain occurs. This is known as a **neural tube** defect.

44. Resting potential membrane?

The opening of the calcium channels involve the receptor's activating a cation channel (for sodium, potassium, or calcium), which would change the resting potential of the sperm plasma membrane. The calcium channels in the membrane would be sensitive to this change in membrane potential, allowing calcium to enter the sperm.

45. Neural plate?

The **neural plate** is a key developmental structure that serves as the basis for the nervous system. Opposite the primitive streak in the embryo, ectodermal tissue thickens and flattens to become the **neural plate**. The region anterior to the primitive knot can be generally referred to as the **neural plate**.

46. Difference between Cytokinesis and Karyokinesis?

Karyokinesis is the process of division of the nucleus of a cell during the phase of division (mitosis or meiosis). **Cytokinesis** is the process of division of cytoplasm of the cell. **Cytokinesis** divides cytoplasm and organelles including nuclei into its daughter cells.

47. Event of Fertilization?

The result of **fertilization** is a cell (zygote) capable of undergoing cell division to form a new individual. The fusion of two gametes initiates several reactions in the egg. One of these causes a change in the egg membrane(s), so that the attachment of and penetration by more than one spermatozoon cannot occur.

48. How Discovered sperm?

A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in 1678, first believed them to be parasitic animals living within the semen (hence the term spermatozoa, meaning "sperm animals").

49. Conditional SP with example?

Conditional specification: o Characteristic of all vertebrates and few invertebrates. o Specification by interactions between cells. Relative positions are important

50. Tunicates gastrulation?

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm. About 4 5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins.

51. Ectoderm difference?

the **ectoderm** differentiates to form the nervous system (spine, peripheral nerves and brain), tooth enamel and the epidermis (the outer part of integument). It also forms the lining of mouth, anus, nostrils, sweat glands, hair and nails

52. MPF Subunit?

Mitosis-promoting factor contains two subunits. The large subunit is called cyclin B. Small subunit of MPF, the cyclin-dependent kinase

53. Defect of Neutralization? (10)

Different neural tube defects are caused when various parts of the neural tube fail to close. Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly

thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed. Failure to close the anterior neural tube regions results in a lethal condition, anencephaly. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates. Fetal forebrain development ceases, and the vault of the skull fails to form.

54. Identical Twin and paternalTwin?

To form **identical** or monozygotic **twins**, one fertilised egg (ovum) splits and develops into two babies with exactly **the same** genetic information. To form **praternal** or dizygotic **twins**, two eggs (ova) are fertilised by two sperm and produce two genetically unique children.

55. Difference mono and Polyspermy?

is that monospermy is fertilization of an oocyte by a single spermatozoon while **polyspermy** is the penetration of an ovum by more than one sperm.

56. What is Syncytiotrophoblast?

The **syncytiotrophoblast** is the primary structure that determines which substances cross the placenta (e.g., nutrients and oxygen) and which substances do not (e.g., maternal hormones and certain toxins).

57. Anatomical Approach?

The third **anatomical approach** to developmental biology is teratology, the study of birth defects. ... The fourth **anatomical approach** is mathematical modeling, which seeks to describe developmental phenomena in terms of equations.

58. Mantal and Ventricular zone?

new layer is called the **mantle** (or intermediate) zone, and the germinal epithelium is now called the **ventricular** zone (and, later, the ependyma).

59. Species Specific recognition in sea urchin?

"**Species specific recognition** of **sea urchin** gametes occurs at the levels of sperm attraction, sperm activation, and sperm adhesion to the egg surface." The **sea-urchin**egg jelly contains a **species specific** chemo-attractant (redact).

60. How avoid polyspermy?

Slow block of **polyspermy**
Nevertheless, **polyspermy preventing** mechanisms are essential in mammals; a secretion reaction, the "cortical reaction" modifies the extracellular coat of the egg (the zona pellucida), and additional mechanisms that are not well understood modify the egg's plasma membrane

61. Chemotoxix?

Is the movement of an organism in response to a chemical

62. Type of Egg in insect?

These are megalecithal polylecithal & centrolecithal **eggs**. The cytoplasm of **insect egg** is limited to periphery only whole central place is filled by yolk.
Two **types of egg** envelops are present on **egg** of **insects**. Inner - Vitelline membrane - Primary **egg** membrane.

63. Two way from neural tube?

There are two major ways of forming a neural tube Primary neurulation Secondary neurulation.

64. Name of cell movement during gastrulation in vertebrates?

During internalization, endoderm and mesoderm **cells** move via the blastopore beneath the ectoderm. ... Patterns of **gastrulation cell movements** relative to the blastopore and the organizer are similar from **fish** to **mammals**, and conserved molecular pathways mediate **gastrulation movements.**

65. Vegetal pole and Animal pole?

The yolk-rich pole is referred to as the **vegetal** pole; the yolk concentration in the **animal** pole is relatively low.

66. Campomelic dysplasia?

Campomelic dysplasia (CMD) is a rare genetic disorder characterized by bowing of the long bones and many other skeletal and extraskeletal features. It is frequently lethal in the neonatal period due to respiratory insufficiency, but the severity of the disease is variable, and some patients survive into adulthood.

67. How egg protected from sunlight?

Many eggs contain ultraviolet filters and DNA repair enzymes that protect them from sunlight

68. Amniotic Fluid?

The amniotic fluid is the protective liquid contained by the amniotic sac of a gravid amniote. This fluid serves as a cushion for the growing fetus, but also serves to facilitate the exchange of nutrients, water, and biochemical products between mother and fetus.

69.What are MFN?

70. What create dramatic change in the structure of organism?

Allometric growth can create dramatic changes in the structure of organisms.

71. Why tunicate give name?

The **name**, "**tunicate**" comes from the firm, but flexible body covering, called a tunic. Most **tunicates** live with the posterior, or lower end of the barrel attached firmly to a fixed object, and have two openings, or siphons, projecting from the other. **Tunicates** are plankton feeders.

72. Transcription factor group together?

General **transcription factors** are involved in the formation of a preinitiation complex. The most common are abbreviated as TFIIA, TFIIB, TFIID, TFIIE, TFIIF, and TFIIH. They are ubiquitous and interact with the core promoter region surrounding the **transcription** start site(s) of all class II genes

73. Difference between Enhancer and Promoter?

An **enhancer** is a sequence of DNA that functions to enhance transcription. A **promoter** is a sequence of DNA that initiates the process of transcription. A **promoter** has to be close to the gene that is being transcribed while an **enhancer** does not need to be close to the gene of interest.

74. Polyspermy avoided in sea urchin?

The most common way is to prevent the entry of more than one sperm into the egg. The **sea urchin** egg has two mechanisms to **avoid polyspermy**: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a slower reaction, caused by the exocytosis of the cortical granules

75. Zona pellucida?

The **zona pellucida** supports communication between oocytes and follicle cells during oogenesis; protects oocytes, eggs, and embryos during development, and regulates interactions between ovulated eggs and free-swimming sperm during and following fertilization.

76. When Capacitation occur?

The set of physiological changes that allow the sperm to be competent to fertilize the egg is called capacitation.

Capacitation is the penultimate step in the maturation of mammalian spermatozoa and is required to render them competent to fertilize an oocyte. ... In vivo, **capacitation occurs** after ejaculation, when the spermatozoa leave the vagina and enter the superior female reproductive tract.

77. What are protostome?

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

78. Composition of hyaline membrane sea urchin?

The **sea urchin** embryo **hyaline** layer is an extracellular investment which develops within 20 min postinsemination of Strongylocentrotus purpuratus eggs and contains a single calcium-precipitable subunit termed hyalin.

79. How neural groove formed in primary neurulation?

The process begins when the notochord induces the **formation** of the central nervous system (CNS) by signaling the ectoderm germ layer above it to **form** the thick and flat **neural plate**. ... In **primary neurulation**, the **neural plate** creases inward until the edges come in contact and fuse.

80. Syncytium?

A cytoplasm that contains many nuclei is called a syncytium

81. Fertilization in Frog?

82. Demonstration of Sperm enter in the egg Two nuclie?

In 1876 Oscar Hertwig and Herman Fol independently demonstrated sperm entry into the egg and the union of the two cells' nuclei.

SUBJECTIVE

2marks:

1. **Define Blastocoel?**

A fluid-filled cavity, the blastocoel, forms in the animal hemisphere. This cavity will be important for allowing cell movements to occur during gastrulation.

2. **Define periosteum?**

The entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the **Periosteum** (a membrane that surrounds the bone).

3. **Define Germ ring?**

After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**, and it is composed of a superficial layer, the **epiblast**, and an inner layer, the **hypoblast**.

4. **What is primitive knot or hensen node?**

At the anterior end of the primitive streak is a regional thickening of cells called the **primitive knot** or **Hensen's node.**

5. **Write difference between Bicoid and Nanos?**

The anterior most portion of the egg contains an mRNA that encodes a protein called Bicoid. The posterior most portion of the egg contains an mRNA that encodes a protein called Nanos.

6. **Difference between axoneme & dyenin?**

Answer: The major motor portion of the flagellum is called the **axoneme.**

The force for sperm propulsion is provided by **dynein**, a protein that is attached to the microtubules.

7. **Define Silencers?**

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

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8. **What is embryonic shield?**

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

9. **What is naïve?**

An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

10. **Egg jelly formation?**

Many types of eggs have glycoprotein meshwork called egg jelly outside the vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

11. **Differentiate between Blastomere and cleavage?**

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called **blastomeres.**

12. **What is marginal belt?**

Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

13. **Define Neural Crest?**

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.

14. **Define Regulative development?**

conditional specification gives rise to a pattern of embryogenesis called **regulative development**.

15. **Neural plate in tunicats?**

In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

16. **Role.of TGF nd FGF7 in human body development**

Paracrine factors such as **TGF- and FGF7** are important in normal skin development.

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17. **Define primary and secondary neurulation?**

In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. **Neurulation in fishes is exclusively secondary.**

18. **Define Prechordal plate?**

Cells migrating through Hensen's node become chordamesoderm (notochord) cells. These extend up to the presumptive midbrain, where they meet the prechordal plate. The **prechordal plate** induces the formation of the forebrain.

3MARKS

19. **Write three distinct lineage of Osteogenesis? Or What are lineages to generate skeleton?**

There are three distinct lineages that generate the skeleton.

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.

20. **What are syncytiotrophoblast and its function?**

The multinucleated type of cell forms the **syncytiotrophoblast**.

The extra-embryonic mesoderm joins the trophoblastic extensions and gives rise to the blood vessels that carry nutrients from the mother to the embryo. The narrow connecting stalk of extra-embryonic mesoderm that links the embryo to the trophoblast eventually forms the vessels of the **umbilical cord**.

21. **Write the function of Metencephalon?**

The Myelencephalon eventually becomes the **Medulla oblongata**, whose neurons generate the nerves that regulate respiratory, gastrointestinal, and cardiovascular movements. The **Metencephalon** gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

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22. Pattern of embryonic cleavage?

Patterns of embryonic cleavage

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. The amount and distribution of yolk protein within the cytoplasm.
2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.
3. The yolk-rich pole is referred to as the **vegetal pole**; the yolk concentration in the **animal pole** is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

23. Write the role of calcium in the fusion of egg and sperm?

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

24. Differentiate between Autonomous and syncytial specification?

Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.
- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.

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Syncytial specification:

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.
- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

25. Difference between evolutionary embryology and tetralogy?

Evolutionary embryology:

The second strand, based on the first, is evolutionary embryology, the study of how changes in development may cause evolutionary changes and of how an organism's ancestry may constrain the types of changes that are possible.

Teratology:

The third anatomical approach to developmental biology is **teratology**, the study of birth defects. These anatomical abnormalities may be caused by mutant genes or by substances in the environment that interfere with development. The study of abnormalities is often used to discover how normal development occurs.

26. Difference b/w identical and fraternal twins?

Identical twins occur in roughly **0.25%** of human births. About **33%** of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at **day 5.**

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

27. Differnce b/w gray matter and white matter?

Glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the **Gray matter**;

The axonal, marginal layer is often called the **White matter**. In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

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28. Define Medial hinge point?

In birds and mammals, the cells at the midline of the neural plate are called the **medial hinge point** (**MHP**) **cells**. They are derived from the portion of the neural plate just anterior to Hensen's node and from the anterior midline of Hensen's node.

29. Diff b/w late and early response of egg to sperm?

Early responses: Contact b/w sea urchin sperm & egg activates the 2 major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, & the slow block, initiated by the intracellular release of Ca+ ions. The activation of all eggs appears to depend on an increase in the conc. of free Ca+ ions within the egg. Such an increase can occur in 2 ways: Ca+ ions can enter the egg from outside, or Ca+ ions can be released from the endoplasmic reticulum within the egg.
Late responses: Shortly after the Ca+ ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a 2nd influx of Na+ ions, which causes a 1:1 exchange b/w sodium ions from the seawater & hydrogen ions from the egg. This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

30. Write the general concept of fertilization?

Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Fertilization accomplishes two separate processes:

- Sex (the combining of genes derived from the two parents) and
- Reproduction (the creation of new organisms)
- Thus, the first function of fertilization is to transmit genes from parent to offspring, and the second is to initiate in the egg cytoplasm those reactions that permit development to proceed.

31. What is Marble bone disease?

Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone to breakage (fracture).

This disorder is characterized by increased bone density and abnormal bone growth.

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32. Estrogen and progesterone function?

Estrogen is a hormone that can instruct the liver to make and secrete the yolk proteins, which are then transported through the blood into the enlarging eggs in the ovary. The yolk is transported into the bottom portion of the egg.

progesterone, signals the egg to resume its meiotic division. This is necessary because the egg had been "frozen" in the metaphase of its first meiosis. When it has completed this first meiotic division, the egg is released from the ovary and can be fertilized.

33. What are resact and its function?

Answer: **Resact** is specific for ***A. punctulata*** and does not attract sperm of other species. *A. punctulata* sperm have receptors in their plasma membranes that bind resact and can swim up a concentration gradient of this compound until they reach the egg. Resact also acts as a **sperm-activating peptide**.

5MARKS:

34. Write the functional domains of neural crest?

- A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.
- The neural crest cells originate at the dorsal most region of the neural tube.
- The neural crest can be divided into four main functional (but overlapping) domains.
- The cranial (cephalic) neural crest, whose cells migrate dorso-laterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia, and connective tissues of the face.

35. life cycle of chlamydomonas?

the life cycle of *Chlamydomonas*. Which is usually haploid, having just one copy of each chromosome. The individuals of each species, however, are divided into two mating types: plus and minus. When a plus and a minus meet, they join their cytoplasm, and their nuclei fuse to form a diploid zygote. The flagella of two individuals twist around each other, enabling specific regions of the cell membranes to come together. These specialized regions contain mating type-specific components that enable the cytoplasms to fuse.

36. Describe primary neurulation?

Primary neurulation

- During primary neurulation, the original ectoderm is divided into three sets of cells.
- The internally positioned neural tube, which will form the brain and spinal cord.

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- The externally positioned epidermis of the skin.
- The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

37. Transcription domain?

Transcription factors have three major domains.

1. **DNA-binding domain**
2. **Protein-protein interaction domain**
3. ***Trans*-activating domain**

- **DNA-binding domain:**It recognizes a particular DNA sequence.
- **Protein-protein interaction domain:** It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.
- ***Trans*-activating domain:** It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this *trans*-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.Examples,***MITF:(***Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.
- ***Pax6:*** Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains.
- **Pdx1:** Pancreatic And Duodenal Homeobox 1 gene. It is specific for the pancreatic region of the endoderm.

38. 2 ways for the formation of neural tube?

Formation of the Neural Tube

There are two major ways of forming a neural tube

1) Primary neurulation 2) **Secondary neurulation**

- In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

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- In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. **Neurulation in fishes is exclusively secondary.**
- In **amphibians**, such as ***Xenopus,*** most of the tadpole neural tube is made by **primary neurulation**, but the tail neural tube is derived from **secondary neurulation**.

39. Enlist 4 factors that cause movement in gene regulation?

The regulation of gene expression can be accomplished at several levels:

- **Differential gene transcription:** It regulates that which of the nuclear genes are transcribed into RNA.
- **Selective nuclear RNA processing:** It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.
- **Selective messenger RNA translation:** It regulates that which of the mRNAs in the cytoplasm become translated into proteins.
- **Differential protein modification:** It regulates that which proteins are allowed to remain or function in the cell.

40. Write modes of commitment?

Three basic modes of commitment are

1: Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.
- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.

Produces "mosaic" ("determinative") development: cells cannot change fate if a blastomere is lost.

2: Conditional specification:

- Characteristic of all vertebrates and few invertebrates.

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- Specification by interactions between cells. Relative positions are important.
- Variable cleavages produce no invariant fate assignments to cells.
- Massive cell rearrangements and migrations precede or accompany specification.

Capacity for "regulative" development: allows cells to acquire different functions.

3: Syncytial specification:

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.
- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

41. Define : Area opaca , Area pellicuda , sub-germinal cavity, marginal zone?

- Between the blastoderm and the yolk is a space called the **sub-germinal cavity**.
- The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the **area opaca**.
- The deep cells in the center of the blastoderm are shed and die, leaving behind a onecell- thick **area pellucida**.
- Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

42. Hammering experiment for role of nucleus in morphology?

Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in *Acetabularia.* This enormous single cell (2-4 cm long) consists of three parts:

- a cap,
- a stalk,
- a rhizoid.

The single nucleus of the cell resides within the rhizoid. The size of *Acetabularia* and the location of its nucleus allow investigators to remove the nucleus from one cell and replace it with a nucleus from another cell.

J.Hammerling exchanged nuclei between two morphologically distinct species, *A. mediterranea* and *A. crenulata* **J. Hammerling** found that when he transferred the nucleus from one species into the stalk of another species, the newly

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formed cap eventually assumed the form associated with the *donor* nucleus. Thus, the nucleus was seen to control *Acetabularia* development.

The formation of a cap is a complex morphogenic event involving the synthesis of numerous proteins, which must be accumulated in a certain portion of the cell and then assembled into complex, species-specific structures.

10 Marks:

43. Endochondral?

- Endochondral ossification involves the formation of cartilage tissue from aggregated mesenchymal cells, and the subsequent replacement of cartilage tissue by bone.
- The process of Endochondral ossification can be divided into five stages.
- First, the mesenchymal cells are committed to become cartilage cells. This commitment is caused by paracrine factors that induce the nearby mesodermal cells to express two transcription factors, Pax1 and **Scleraxis**.
- Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes.
- Thus, Scleraxis is expressed in the mesenchyme from the sclerotome, in the facial mesenchyme that forms cartilaginous precursors to bone, and in the limb mesenchyme
- During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells. N-cadherin appears to be important in the initiation of these condensations.
- During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells
- In humans, the *SOX9* gene, which encodes a DNA-binding protein, is expressed in the precartilaginous condensations. Mutations of the *SOX9* gene cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body.
- During the third phase of endochondral ossification, the chondrocytes proliferate rapidly to form the model for the bone. As they divide, the chondrocytes secrete a cartilage-specific extracellular matrix In the fourth phase, the chondrocytes stop dividing and increase their volume dramatically, becoming hypertrophic chondrocytes. These large chondrocytes alter the matrix they produce (by adding collagen X and more fibronectin) to enable it to become mineralized by calcium carbonate.
- The fifth phase involves the invasion of the cartilage model by blood vessels.
- **The hypertrophic chondrocytes die by apoptosis.** This space will become **bone marrow**. As the cartilage cells die, a group of cells that have surrounded the cartilage model differentiate into osteoblasts.
- The osteoblasts begin forming bone matrix on the partially degraded cartilage. Eventually, all the cartilage is replaced by bone. Thus, the cartilage tissue serves as a model for the bone that follows

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44. Explain Osteogenesis in detail?

Osteogenesis:

- Some of the most obvious structures derived from the paraxial mesoderm are bones.

There are three distinct lineages that generate the skeleton.

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.
- There are two major modes of bone formation, or **osteogenesis**, and both involve the transformation of a preexisting mesenchymal tissue into bone tissue.
- The direct conversion of mesenchymal tissue into bone is called **intramembranous ossification**. This process occurs primarily in the bones of the skull.
- In other cases, the mesenchymal cells differentiate into cartilage, and this cartilage is later replaced by bone. The process by which a cartilage intermediate is formed and replaced by bone cells is called endochondral ossification.

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